

ADAPTIVE RECOVERABILITY GOVERNANCE

A Supervisory Layer Above VaR, Stress Testing, and Catastrophe Modelling

Nicholas D'Zilva

Abstract

Modern financial risk systems — Value at Risk, Stressed VaR, and catastrophe models — are highly effective at estimating loss exposure under defined scenarios. They answer the question: what losses may occur?

They do not answer a prior and more fundamental question: can this institution still recover using its own internal correction mechanisms?

This paper introduces Adaptive Recoverability Governance (ARG): a supervisory overlay designed to sit above existing risk infrastructure. ARG integrates two components:

- The Zombie Sentinel Layer — monitoring latent deterioration in institutional correction capacity through four auditable structural dimensions and a perceptual convergence monitor.
- The Decision Robustness Layer — dynamically constraining exposure, leverage, and irreversible commitments as measured recoverability declines.

Rather than replacing existing systems, ARG acts as a state-aware governance layer that responds to structural health signals before visible instability emerges. The framework is sector-agnostic and has been validated computationally across banking, reinsurance, and sovereign risk profiles.

1. Existing Risk Frameworks and Their Shared Limitation

1.1 Value at Risk

VaR estimates the maximum expected loss over a defined period at a given confidence interval under normal market conditions. It is effective for market exposure measurement, capital allocation, and day-to-day portfolio monitoring.

VaR assumes relatively stable market structure, functioning liquidity, and — critically — persistent institutional recoverability. The capacity of the institution to correct itself is treated as fixed and given.

1.2 Stressed Value at Risk

SVaR extends VaR by calibrating to a historically stressed period, improving tail-risk estimation and regulatory resilience. However, SVaR still assumes that institutions remain governable and that correction mechanisms continue functioning under stress. It tests what the exposure would be during a historical crisis; it does not test whether the institution retains the structural capacity to respond.

1.3 Catastrophe Models

CAT models simulate low-frequency, high-severity events across natural, cyber, pandemic, and systemic scenarios. They are highly effective for exposure aggregation, scenario modelling, and loss estimation. However, they primarily evaluate event severity and financial loss transmission. They do not model endogenous governance degradation, recoverability decay, or adaptive institutional constraint escalation.

1.4 The Shared Limitation

Every tool above shares a foundational assumption: the system being measured is structurally stable and capable of self-correction. The question being asked is always some form of how bad could a bad event be — never whether the institution is still capable of recovering when it does get bad.

This gap is not theoretical. It is the consistent structural pattern across every major institutional failure of the past thirty years:

- Japanese banking system, 1990s–2000s: banks passed contemporaneous solvency metrics while correction capacity had been structurally absent for years.
- Lehman Brothers, 2008: internal risk models, VaR, and stress test outputs remained within accepted ranges in the quarters before collapse.
- Silicon Valley Bank, 2023: capital ratios were compliant; the supervisory framework did not detect that the institution's correction mechanisms — including the oversight relationship with its primary regulator — were structurally compromised.

In each case, the institution entered what this framework calls the zombie zone: a regime in which the system continues to operate and produce outputs while having lost the endogenous capacity to self-correct. The failure moment was not the visible collapse — it was the earlier, undetected entry into that regime.

2. The Missing Layer: Recoverability

The central distinction this framework introduces is between structural persistence and regulatory functionality.

An institution may remain solvent, operational, profitable, and externally stable while internally:

- governance quality deteriorates,
- optionality shrinks,
- intervention costs escalate nonlinearly,
- and endogenous recovery capacity weakens.

This creates a condition of operational persistence with declining recoverability. Existing systems will not detect it because they are not designed to measure it. ARG is.

Core proposition: *The most important moment in an institution's risk lifecycle is not collapse. It is the moment the institution enters a state where recovery is no longer endogenous — where continued functioning depends on external support rather than internal correction capacity.*

The framework measures recoverability directly through two integrated layers described in the sections that follow.

3. The Zombie Sentinel Layer

3.1 Structural State Classification

The Zombie Sentinel classifies institutional state across three regimes, defined by the relationship between raw distortion intensity (DI) and correction capacity (C):

- Functional: $C \geq DI$. The institution can self-correct. Internal correction mechanisms are intact.
- Zombie Zone: $C < DI$ and $C > 0.1$. The institution persists structurally but has lost endogenous correction capacity. Self-correction becomes progressively less feasible without external intervention. Intervention cost escalates nonlinearly with depth.
- Collapse: $C \leq 0.1$. Structural failure. Correction capacity is effectively absent.

The critical insight is that entry into the zombie zone — not collapse — is the sentinel event. After that transition, the institution's trajectory is determined by time and external intervention, not internal dynamics.

3.2 The Correction Integrity Index (CII)

Conventional frameworks treat correction capacity as fixed and implicit. The CII makes it explicit, measurable, and auditable. C is computed as a weighted composite of four observable institutional dimensions, then adjusted for observation capture:

Component	What It Measures	Financial Proxies	Score Range
Supervisory Independence (SI)	Whether oversight bodies can act free from institutional or financial dependency on supervised entities	Enforcement action rate; regulator funding source; revolving-door tenure density	1.0 = fully independent 0.1 = structurally captured
Analytical Diversity (AD)	Whether risk assessment draws on genuinely independent frameworks, not converged consensus models	Model concentration (HHI); rating agency input diversity; independent challenge frequency	1.0 = active challenge 0.1 = single-model monoculture
Contestability (CO)	Whether genuine competitive or regulatory challenge would force correction if internal mechanisms fail	Sector HHI; credible resolution regime; too-big-to-fail designation; bail-out probability	1.0 = genuine contestability 0.1 = systemically protected
External Permeability (EP)	Whether the institution is open to corrective signals from outside its own incentive structure	External audit tenure; whistleblower outcomes; Pillar 3 transparency; independent research coverage	1.0 = active external challenge 0.1 = opaque, long-tenured auditor

Observation Capture (OC) is applied as a penalty multiplier (0.0 to 0.40) on the composite CII score. OC is distinct from regulatory forbearance — which is a policy choice — and describes a structural condition where the bodies responsible for detecting failure are themselves institutionally dependent on the entities they oversee. SVB’s primary regulator operated under conditions of elevated observation capture. The framework identifies and quantifies this condition explicitly.

Final correction capacity: $C = CII \text{ composite} \times (1 - OC \text{ penalty})$

3.3 The Perceptual Convergence Monitor (PCM)

Correction capacity does not decay randomly. It decays through a specific and measurable process: the progressive synchronisation of institutional perception under shared incentive environments.

When institutional actors — risk officers, regulators, rating analysts, board members — operate within the same adaptive incentive environment, they progressively lose interpretive divergence. Career incentives synchronise. Acceptable analytical frameworks narrow. Correction pathways become professionally illegible. This is the mechanism that produces analytical monoculture and destroys the independent error-checking that genuine correction capacity requires.

The PCM monitors four convergence indicators with defined financial proxies:

Convergence Indicator	Financial Manifestation	Measurement Proxy
-----------------------	-------------------------	-------------------

Rating Agency Clustering	All major agencies move together, eliminating independent signal value	Pairwise correlation of S&P, Moody's, Fitch rating changes over 24 months
Model Concentration	Industry-wide adoption of identical vendor models eliminates independent error-checking	HHI of risk models in use; vendor market share concentration
Board Homogeneity	Risk committee positions converge toward industry consensus rather than independent analysis	Background diversity index of risk committee members; institution-type tenure concentration
Regulatory Isomorphism	Regulators adopt the same analytical frameworks as supervised institutions, removing independent verification	Framework similarity score between internal models and regulatory stress scenarios

The composite Convergence Score (CS) applies a decay multiplier to C: $C_{\text{adjusted}} = C \times (1 - 0.30 \times \text{CS})$. An institution with a CS of 0.70 — typical of severe analytical monoculture — has its correction capacity reduced by 21% from convergence alone, before any external shock occurs.

3.4 Zombie Zone Depth, Velocity, and Acceleration

Once an institution enters the zombie zone, three metrics characterise its trajectory:

- δ (zombie depth): how far the institution has progressed from zombie zone entry toward collapse. $\delta = 0$ is entry; $\delta = 1$ is collapse boundary.
- $d\delta/dt$ (velocity): the rate of deterioration. Positive velocity indicates active deepening.
- $d^2\delta/dt^2$ (acceleration): the rate of change of velocity. Positive acceleration indicates runaway deterioration — the leading indicator for imminent governance escalation.

These three metrics are leading indicators unavailable in any existing risk framework. VaR and stress testing are contemporaneous or scenario-based. Zombie depth, velocity, and acceleration describe the structural trajectory — what the risk picture will be if current dynamics continue.

The Allee threshold introduces a fourth and distinct diagnostic: nonlinear collapse detection. When effective correction capacity decays below a critical threshold (C_{crit}), the correction mechanism does not continue degrading smoothly — it flips. Effective distortion spikes discontinuously as the Theta function switches from operative ($\Theta = 1$) to collapsed ($\Theta = 0$). This models the failure pattern observed in SVB, Lehman, and the Japanese banking system: prolonged gradual deterioration followed by sudden, discontinuous collapse that conventional metrics did not anticipate. The governance layer is calibrated to reach maximum constraint before C_{crit} is crossed — because after the flip, there is no gradual response window.

4. The Decision Robustness Layer

4.1 The Weighted Divergence Score

The Decision Robustness Layer aggregates all structural and market signals into a single Weighted Divergence Score (WDS). Nine signal channels feed the WDS engine:

- Zombie depth (δ), velocity, and acceleration — structural health signals from the Zombie Sentinel.
- Market stress — credit spread deviation, asset price divergence from fundamental value.
- Claims velocity — claims rate relative to historical baseline.
- Macro stress — rates, GDP trajectory, foreign exchange volatility.
- CII score — correction integrity signal (inverted: low CII = high stress).
- Convergence score — perceptual convergence level from the PCM.
- Observation capture — degree of regulator and auditor capture.

WDS applies confidence-weighted divergence scoring across all channels, with an additional coupling multiplier that activates when previously independent signals begin moving together. Correlated signal movement is the primary regime-change indicator: it means the system is no longer experiencing isolated stress but structural co-movement. The coupling penalty reflects this: $WDS_{adjusted} = WDS \times (1 + 0.2 \times \text{coupled_pairs})$.

Confidence weights decay asymmetrically under high WDS conditions — decaying rapidly (50% per period above WDS 0.6) and recovering slowly (requiring three consecutive stable periods). This prevents premature relaxation of governance constraints during volatile periods.

4.2 Governance Constraint Escalation

Governance constraint strength (G) is a sigmoid function of the adjusted WDS:

$$G = 1 / (1 + e^{(-12 \times (WDS - 0.35))})$$

This produces four governance regimes:

- LOW (WDS < 0.3): Correlations stable. Near-unconstrained commitment authority.
- MODERATE (WDS 0.3–0.6): Monitoring increased. Exposure envelopes begin contracting.
- HIGH (WDS 0.6–0.8): Envelopes tighten materially. Staged commitment tranches required.
- EXTREME (WDS > 0.8): Maximum constraint activation. New commitments effectively halted.

Three invariants operate as hard governance floors regardless of WDS level: no single commitment may exceed 10% of capital (Bounded Exposure); liquidity must remain above 25% of capital post-deployment (Preserved Liquidity); under high WDS, commitments are split into immediate and reserved tranches pending resolution (Maintained Optionality).

Crucially, governance tightens in response to structural health signals — not in response to trigger events. The system does not wait for a crisis threshold to be breached. It contracts exposure as structural decay is detected.

5. Relationship to Existing Risk Systems

ARG is not intended to replace VaR, SVaR, CAT models, liquidity systems, or regulatory capital frameworks. These systems continue to perform the functions they are designed for: loss estimation, capital allocation, and scenario analysis.

ARG acts as a supervisory governance and recoverability overlay. The relationship is additive:

Existing Risk Systems	Adaptive Recoverability Governance
What is the exposure?	Can the institution still correct itself?
How bad could a bad event be?	Is the institution structurally equipped to respond when it does?
Event-driven: responds after threshold breach	State-aware: responds to structural decay trajectory
Contemporaneous or scenario-based signals	Leading indicators: depth, velocity, acceleration
C assumed constant and implicit	C measured, tracked, and decomposed into auditable components
Convergence and capture not modelled	Perceptual convergence and observation capture quantified

Institutions already running VaR, SVaR, and CAT systems can integrate ARG as an additional layer without modifying existing infrastructure. The nine WDS signal channels can be fed from existing data pipelines.

6. Computational Validation

The unified framework has been implemented as a working computational system and validated across three institution profiles. All three scenarios were run with identical shock parameters (6% monthly shock probability, fixed random seed) to isolate the effect of structural health differences.

Institution Profile	CII Score (C)	Convergence Score	Effective C	Outcome
Strong: High SI, AD, CO, EP; low convergence	0.76	0.19	0.71	Survives 100+ periods; governance active throughout
Degraded: Weak oversight, high convergence, 25% OC penalty	0.30	0.59	0.20	Collapse at period 12; EXTREME WDS flagged at period 8
SVB-style: Critical CII, severe convergence, 35% OC, 30% illusion penalty	0.20	0.70	0.11	Collapse at period 4; WDS maxes at 1.0 immediately

The SVB-style scenario is particularly instructive. With a CII of 0.20, a convergence score of 0.70, and 35% observation capture, the system's effective correction capacity was 0.11 before any shocks occurred. The WDS engine registered EXTREME on the first computation. The governance layer halted new commitments at period 1. The institution collapsed at period 4 — but with capital substantially intact, because the governance layer had already stopped new exposure.

This is the operational proposition: the framework does not prevent external shocks. It ensures that governance posture reflects structural reality before those shocks arrive.

7. Sector Applications

The framework is sector-agnostic. The distortion components (M, R, A, X), CII scoring weights, and PCM indicators can be calibrated to any risk-bearing institution or system. Priority sectors for initial deployment:

- Reinsurance and specialty insurance: parametric governance over catastrophe exposure. WDS detects correlated signal movement before catastrophe events materialise, contracting portfolio exposure before trigger conditions are met.
- Banking and prudential supervision: CII and PCM provide a structured quantitative methodology for supervisory assessment of institutional correction capacity — a dimension currently handled qualitatively under Basel Pillar 2.
- Sovereign and emerging market risk: zombie depth and velocity provide 12–24 month leading indicators of structural deterioration ahead of ratings downgrades and debt distress events.
- Infrastructure and project finance: long-duration exposure makes correction capacity monitoring especially valuable. A 30-year concession with a counterparty in early zombie zone represents materially higher risk than credit ratings reflect.
- Regulatory bodies and central banks: CII provides an auditable framework for assessing institutional governance quality across supervised sectors, with the potential for integration into Basel-equivalent stress testing methodologies.

8. Conclusion

Modern financial systems operate in environments of interconnected fragility, rapid feedback loops, and nonlinear systemic stress. Traditional risk systems remain essential for exposure estimation and capital management.

However, the consistent structural pattern across major institutional failures is not sudden collapse. It is operational persistence with undetected declining recoverability — institutions that continued to function, absorb resources, and pass conventional metrics while having already lost the capacity to correct themselves.

Institution Profile	CII Score (C)	Convergence Score	Effective C	Allee Status	Outcome
Strong: High SI, AD, CO, EP; low convergence	0.76	0.19	0.71	t=100 — C decays to 0.129, Theta flips after 15 zombie periods	Survives 100+ periods; capital intact; governance active throughout
Degraded: Weak oversight, high convergence, 25% OC penalty	0.30	0.59	0.20	t=7 — Allee flip triggers; WDS already at EXTREME before collapse	Collapse at period 7; governance had halted new exposure at period 3
SVB-style: Critical CII, severe convergence, 35% OC, 30% illusion penalty	0.20	0.70	0.103	t=0 — Below C _{ocrit} before simulation begins. System was never recoverable.	Immediate Allee Collapse; no simulation required to determine outcome

Adaptive Recoverability Governance monitors that transition and dynamically adapts governance behaviour before visible instability emerges. It does not compete with existing risk infrastructure. It provides the supervisory layer above it that the industry has not yet had.

Nicholas D'Zilva | Confidential — For Discussion Purposes